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Solving Linear Systems with the Harrow-Hassidim-Lloyd Quantum Algorithm

PhD course "The ABCs of Quantum Computing"

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Harrow et al.2009

Let $Ax = b$ a linear system, $A \in \mathbb{R}^{N_b \times N_b}$, $b \in \mathbb{R}^{N_b}$

- **Input limitations:** A sparse and hermitian
- **Output limitations:** HHL **does not** actually compute x (*amplitude encoding*)

$$|x\rangle = \sum_{j=0}^{N_b-1} x_j |j\rangle$$

  Starting from this: $x^\top Mx$
- **Logarithmic scaling in N_b:**

$$\mathcal{O}(\log N_b \kappa^2/\epsilon) \quad \text{vs} \quad \mathcal{O}(N_b \kappa)$$

  κ: condition number of A, ϵ: desired accuracy

Basic idea

Given the eigenvalue decomposition (EVD) of the matrix, $A = \sum_i \lambda_i u_i u_i^\top$, its inverse can be computed as

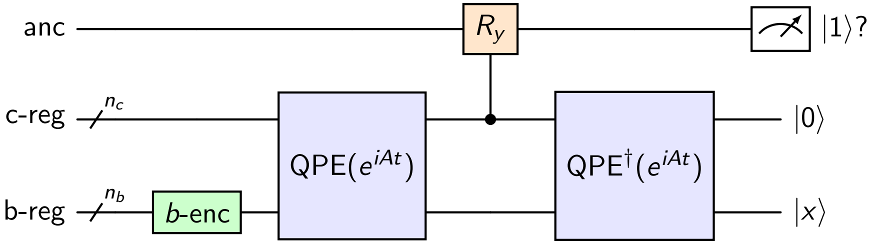
$$A^{-1} = \sum_i \lambda_i^{-1} u_i u_i^\top$$

Thus, if we rewrite b in the eigenvector basis, $b = \sum_i \beta_i u_i$, x will be

$$x = A^{-1} b = \sum_i \lambda_i^{-1} u_i u_i^\top b = \sum_i \lambda_i^{-1} \beta_i u_i$$

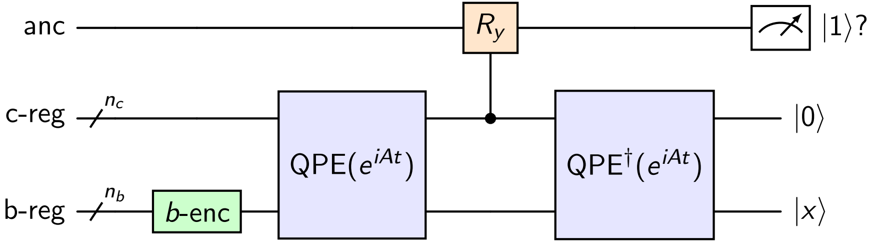
Registers:

- *b-register*: components of b, x, (*amplitude encoding*), n_b qubits, with $N_b = 2^{n_b}$
- *c-register*: phase of the eigenvalues of A, n_c qubits, $N_c = 2^{n_c}$
- *ancilla*: inversion and measurement



Initial state:

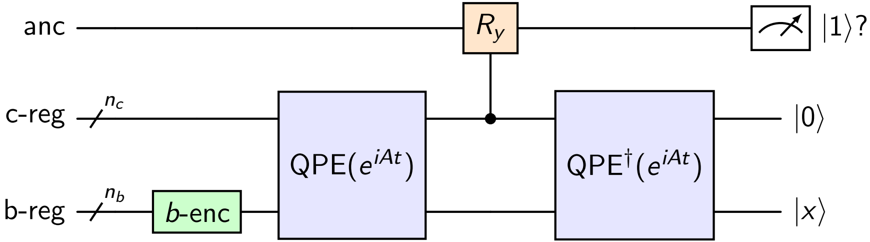
$$|\Psi_0\rangle = |0\rangle^{\otimes n_b} |0\rangle^{\otimes n_c} |0\rangle$$



HHL implementation

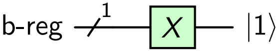
Steps:

1. State preparation
2. QPE
3. Ancilla qubit rotation
4. Inverse QPE



State preparation: rotate the qubits in the b-register to have amplitudes corresponding to the coefficients of b (*amplitude encoding*)

A simple example: $b = (0, 1), N_b = 2 \implies n_b = 1$



Recall:

- state preparation step specific to each b
- b needs to be normalized: $\sum_i |\tilde{b}_i|^2 = 1$
- output: $|\Psi_0\rangle = |b\rangle |0\rangle^{\otimes n_c} |0\rangle$

Quantum Phase Estimatation (QPE)

Goal: estimate the phase of the eigenvalues of A

Issues:

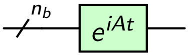
- How to encode matrix A in the circuit? *Hamiltonian encoding*
- How to estimate its eigenvalues? *Quantum Phase Estimation*

Hamiltonian encoding

In general, A non-unitary: we'll use

$$U = e^{iAt}$$

which is unitary and can be approximated by solving a *Hamiltonian simulation* problem



Problem (Hamiltonian Simulation Problem)

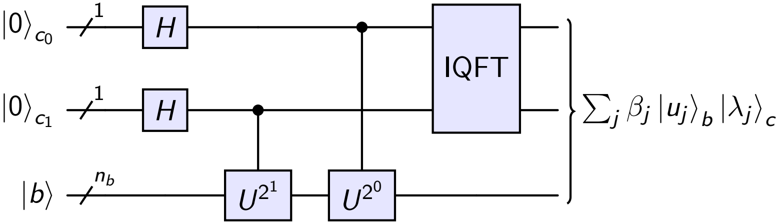
Given a Hamiltonian ($2^{n_b} \times 2^{n_b}$ Hermitian matrix), A, a time t and a tolerance ϵ, find U such that

$$||U - e^{iAt}|| \leq \epsilon$$

Algorithm: Trotterization + Pauli gadget ($A = \sum_j A_j$, $U \approx \prod_j e^{iA_j t}$)

Quantum Phase Estimation (QPE)

Eigenvalue connection: $\mathrm{eig}(U) = e^{i\lambda_j t}$, where λ_j eigenvalues of A



Steps:

1. Superposition
2. Controlled rotations
3. Inverse Quantum Fourier Transforms (IQFT)

First step: superimpose the qubits:

$$|\Psi_2\rangle = |b\rangle \frac{1}{2^{\frac{n_c}{2}}}(|0\rangle + |1\rangle)^{\otimes n} |0\rangle_a$$

This is a prerequisite to exploit *phase kickback* in subsequent steps

QPE: Controlled Rotations I

For the moment, let's assume $|b\rangle$ eigenvector of U with eigenvalue $e^{i2\pi\phi}$:

$$U|b\rangle = e^{i2\pi\phi}|b\rangle$$

Recall that we can always express $|b\rangle$ in terms of the basis of eigenvectors of U

$$|b\rangle = \sum_j \beta_j |u_j\rangle$$

If the control is 0, $|b\rangle$ is not affected, otherwise U is applied. This is equivalent to (*phase kickback*):

$$
\begin{aligned}
|\Psi_3\rangle &= (\frac{1}{2^{\frac{n_c}{2}}}(|b\rangle|0\rangle + U^{2^{n_c}-1}|b\rangle|1\rangle) \otimes \cdots \otimes (|0\rangle + U^{2^0}|b\rangle|1\rangle)) \otimes |0\rangle_a \\
&= |b\rangle \otimes (\frac{1}{2^{\frac{n_c}{2}}}(|0\rangle + e^{i2\pi\phi(2^{n_c}-1)}|1\rangle) \otimes \cdots \otimes (|0\rangle + e^{i2\pi\phi 2^0}|1\rangle)) \otimes |0\rangle_a \\
&= |b\rangle \otimes (\frac{1}{\sqrt{N_c}}\sum_{k=1}^{N_c-1} e^{i2\pi\phi k}|k\rangle) \otimes |0\rangle_a
\end{aligned}
$$

Motivation: transfer phase information from U to the *c-registers*!

QFT maps a **state in the computational basis to a state in the Fourier basis**. Let $|x\rangle = \sum_{j=0}^{N_c} x_j |j\rangle$ expressed in the computational basis

$$|j\rangle \rightarrow U_{\mathrm{QFT}} |j\rangle = \frac{1}{\sqrt{N_c}} \sum_{k=0}^{N_c} e^{i2\pi jk/N_c} |k\rangle$$

leading to $\mathrm{QFT}(|x\rangle) = \sum_j x_j QFT(|j\rangle)$ expressed in the fourier basis. Unitary matrix representation:

$$U_{\mathrm{QFT}} = \frac{1}{\sqrt{N_c}} \begin{bmatrix} 1 & \dots & 1 \\ 1 & \dots & e^{i2\pi(N_c-1)/N_c} \\ \vdots & \ddots & \vdots \\ 1 & \dots & e^{i2\pi(N_c-1)(N_c-1)/N_c} \end{bmatrix}$$

QPE: Inverse Quantum Fourier Transform

Observe that

$$|\Psi_3\rangle = |b\rangle \otimes \Big(\frac{1}{\sqrt{N_c}} \sum_{k=0}^{N_c-1} e^{i2\pi\phi k} |k\rangle\Big) \otimes |0\rangle_a$$

stores phase information ($|N_c\phi\rangle$) in the Fourier basis. Indeed,

$$U_{\mathrm{QFT}} |N_c\phi\rangle = \frac{1}{\sqrt{N_c}} \sum_{k=0}^{N_c-1} e^{i2\pi N_c\phi k/N_c} |k\rangle$$

Thus, $\mathrm{IQFT}(\mathrm{QFT}(|N_c\phi\rangle)) = |N_c\phi\rangle$

Recap:

- We assumed: $|b\rangle$ eigenvector of U with eigenvalue $e^{i2\pi\phi}$
- U encodes A as $U = e^{iAt}$, so that $U|b\rangle = e^{i\lambda_j t}|b\rangle$

Thus, $\phi = \frac{\lambda_j t}{2\pi}$ and

$$|\Psi_4\rangle = |u_j\rangle \left|\frac{N_c \lambda_j t}{2\pi}\right\rangle |0\rangle_a$$

More generally, writing $|b\rangle = \sum_j \beta_j |u_j\rangle$

$$|\Psi_4\rangle = \sum_{j=0}^{2^{n_b}-1} \beta_j |u_j\rangle \left|\frac{N_c \lambda_j t}{2\pi}\right\rangle |0\rangle_a$$

Ancilla qubit

Define $\tilde{\lambda}_j = \frac{N_c \lambda_j t}{2\pi}$

$$|\Psi_5\rangle = \sum_{j=0}^{2^{n_b}-1} \beta_j \, |u_j\rangle \, |\tilde{\lambda}_j\rangle \, (\sqrt{1 - \frac{C^2}{\tilde{\lambda}_j^2}} \, |0\rangle_a + \frac{C}{\tilde{\lambda}_j} \, |1\rangle_a)$$

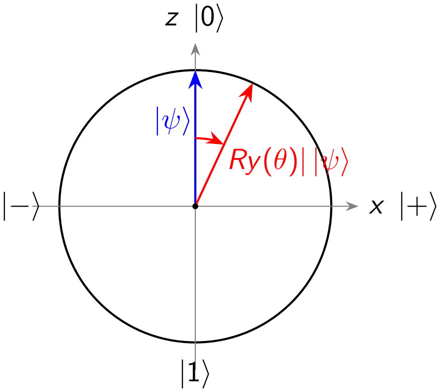
Rotation of the ancilla qubit: eigenvalue inversion, depends on a parameter C.

Ry-gates:

$$R_y(\theta_j) = \begin{bmatrix} \cos(\theta_j/2) & -\sin(\theta_j/2) \\ \sin(\theta_j/2) & \cos(\theta_j/2) \end{bmatrix}$$

and angle

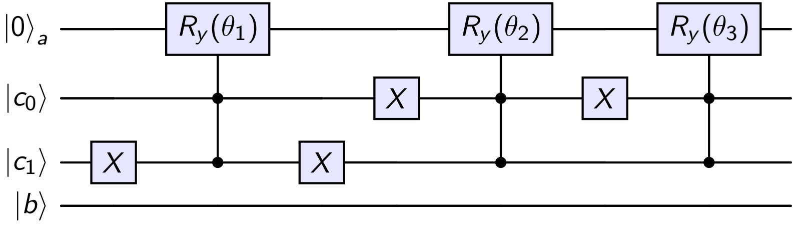
$$\theta_j = \arcsin(\frac{C}{\tilde{\lambda}_j})$$



Ancilla Qubit

Implementation: Map each possible state of the *c-register* to $|1\dots1\rangle$ and relate it to its specific rotation.

Before rotation: $\sum_j \beta_j \, |u_j\rangle_b \, |\tilde{\lambda}_j\rangle_c \, |0\rangle_a$



Only the rotations related to the encoded $\tilde{\lambda}_j$ will trigger.

Ancilla qubit

Once the ancilla qubit is measured we got two alternatives

- if $|0\rangle$: throw away the results
- if $|1\rangle$: store the results

After that the state is updated to

$$|\Psi_6\rangle = \frac{1}{\sum_{j=0}^{2^{n_b}-1} |\frac{\beta_j N_c}{\tilde{\lambda}_j}|^2} \sum_{j=0}^{2^{n_b}-1} \frac{C}{\tilde{\lambda}_j} \beta_j |u_j\rangle |\tilde{\lambda}_j\rangle |1\rangle_a$$

Three hyperparameters: (n_c, t, C). To set them:

- (t, n_c) determine a discrete grid to estimate the eigenvalues of A, λ_j
- $|0 \dots 0\rangle$ cannot store any λ_j
- C has to be lower than $\min_j \lambda_j$ but as high as possible

Recall $\phi = \frac{\lambda_j t}{2\pi}$ with $\phi \in [0, 1)$ and that the *c-registers* store $|N_c\phi\rangle$. Then

- **Scale:**
$$\phi < 1 \implies t < \frac{2\pi}{\max_j \lambda_j}$$
- **Sensibility:**
$$\Delta\phi = \frac{1}{N_c} \implies \Delta\lambda = \frac{2\pi}{tN_c}$$

Moreover, the fact that $|0 \dots 0\rangle_c$ cannot store any $\tilde{\lambda}_j$ leads to $t > \frac{2\pi}{N_c \min_j \lambda_j}$ and $N_c > \frac{\max_j \lambda_j}{\min_j \lambda_j} = \kappa$

Inverse QPE

$$|\Psi_6\rangle = \frac{1}{\sum_{j=0}^{2^{n_b}-1} |\frac{\beta_j N_c}{\tilde{\lambda}_j}|^2} \sum_{j=0}^{2^{n_b}-1} C \frac{\beta_j}{\tilde{\lambda}_j} |u_j\rangle |\tilde{\lambda}_j\rangle |1\rangle_a$$

Issue: *b-registers* storing the solution in the desired form are entangled with the *c-registers*

Firstly, QFT is applied to the *c-register* qubits:

$$\mathrm{QFT}(|\tilde{\lambda}_j\rangle) = \frac{1}{2^{\frac{n}{2}}} \sum_{k=0}^{N_c-1} e^{2\pi i k \tilde{\lambda}_j / N_c} |k\rangle$$

leading to

$$|\Psi_7\rangle = \frac{1}{\sum_{j=0}^{2^{n_b}-1} |\frac{\beta_j N_c}{\tilde{\lambda}_j}|^2} \sum_{j=0}^{2^{n_b}-1} C \frac{\beta_j}{\tilde{\lambda}_j} |u_j\rangle \left(\frac{1}{2^{\frac{n}{2}}} \sum_{k=0}^{N_c-1} e^{2\pi i k \tilde{\lambda}_j / N_c} |k\rangle\right) |1\rangle_a$$

Then controlled rotations with U^{-1} are applied as before

$$\sum_{j=0}^{2^{n_b}-1} C\frac{\beta_j}{\tilde{\lambda}_j}\left|u_j\right\rangle \left(\frac{1}{2^{\frac{n}{2}}}\sum_{k=0}^{N_c-1} e^{-i\lambda_j tk} e^{2\pi ik\tilde{\lambda}_j/N_c}\left|k\right\rangle\right)$$

$$=\sum_{j=0}^{2^{n_b}-1} C\frac{\beta_j}{\tilde{\lambda}_j}\left|u_j\right\rangle \sum_{k=0}^{N_c-1}\left|k\right\rangle = C\left|x\right\rangle \sum_{k=0}^{N_c-1}\frac{1}{2^{\frac{n}{2}}}\left|k\right\rangle$$

leading to

$$\left|\Psi_8\right\rangle = \frac{C}{2^{\frac{n}{2}}\sum_{j=0}^{2^{n_b}-1}\left|\frac{\beta_j N_c}{\tilde{\lambda}_j}\right|^2}\left|x\right\rangle \sum_{k=0}^{N_c-1}\frac{1}{2^{\frac{n}{2}}}\left|k\right\rangle\left|1\right\rangle_a$$

The clock-qubits and *b-registers* are now unentagled. By applying the Hadamard gates to the clock qubits we have

$$|\Psi_9\rangle = \frac{C}{\sum_{j=0}^{2^{n_b}-1} |\frac{\beta_j N_c}{\tilde{\lambda}_j}|^2} |x\rangle_b |0\rangle_c^{\otimes n} |1\rangle_a$$

where the solution $|x\rangle$ is stored in the *b-registers* successfully.

Linear system, $Ax = b$

$$A = \begin{bmatrix} 1 & -1/3 \\ -1/3 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

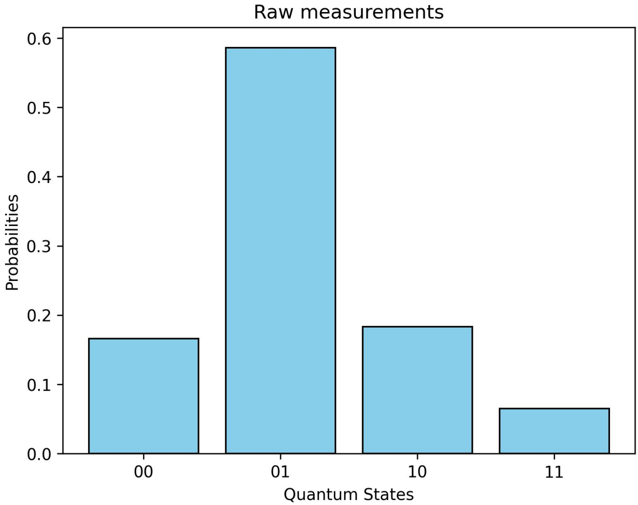
Actual solution

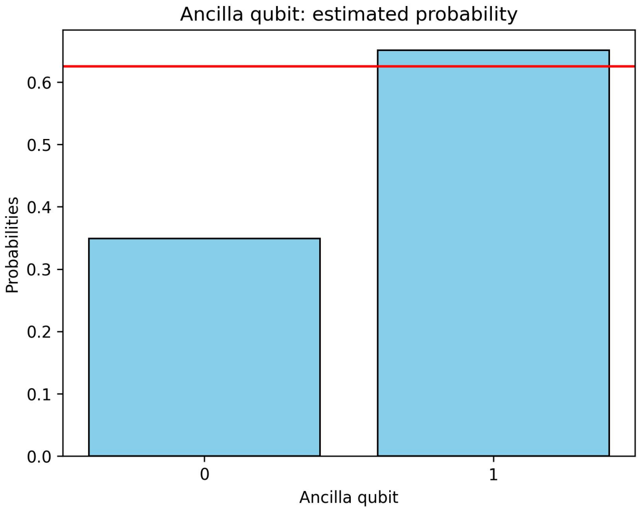
$$x = \begin{bmatrix} 1.125 \\ 0.375 \end{bmatrix}$$

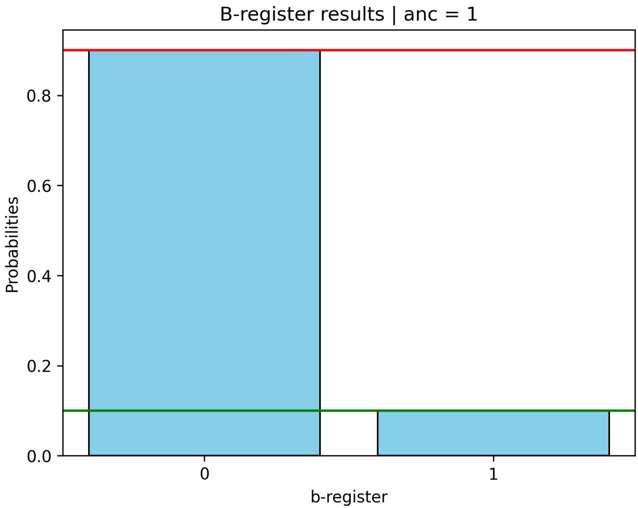
Expected results:

$$\frac{1}{\sum_{j=0}^{1} x_j^2} \begin{bmatrix} x_0^2 \\ x_1^2 \end{bmatrix} = \begin{bmatrix} 0.9 \\ 0.1 \end{bmatrix}, \quad \mathbb{P}(\text{anc} = |1\rangle) = \sum_{j=0}^{1} \left(\frac{C\beta_j}{\lambda_j} \right)^2 = 0.625$$

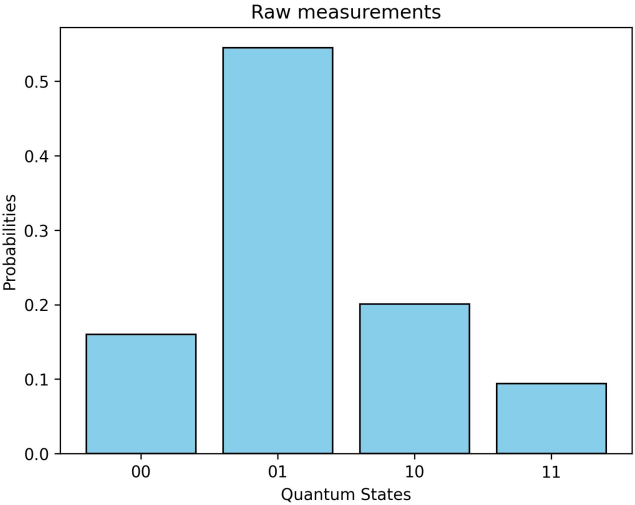
Simulations: Quantum Emulator



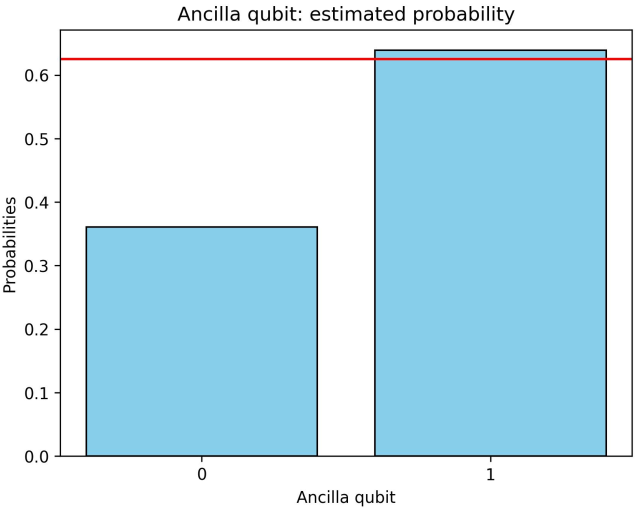

Ancilla qubit: estimated probability
Probabilities
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0
1
Ancilla qubit


B-register results | anc = 1
Probabilities
0.0
0.2
0.4
0.6
0.8
0
1
b-register

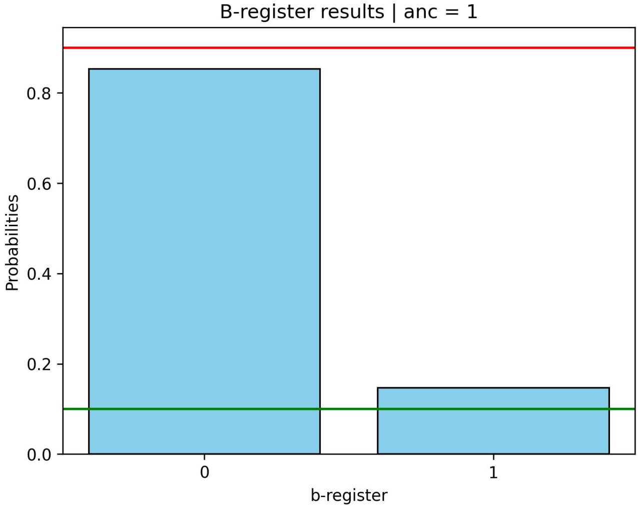
Simulations: IBM Quantum Hardware



Simulations: IBM Quantum Hardware



Simulations: IBM Quantum Hardware



Pros:

- A quantum algorithm for linear systems
- Instructive: several subroutines common to many other quantum algorithms (QFT, QPE, Hamiltonian simulation)

Cons:

- Limited: $x^\top M x$
- Deep circuit: noise on real quantum hardware

Bibliography

A. W. Harrow, A. Hassidim, and S. Lloyd. Quantum algorithm for linear systems of equations. *Physical Review Letters*, 103(15), 2009.

A. Zaman, H. J. Morrell, and H. Y. Wong. A step-by-step HHL algorithm walkthrough to enhance understanding of critical quantum computing concepts. *IEEE Access*, 11:77117–77131, 2023.